

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1501

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Code of Conduct

19251171

I H. H. B. Shewar Khan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: H. H. B. Shewar Khan

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

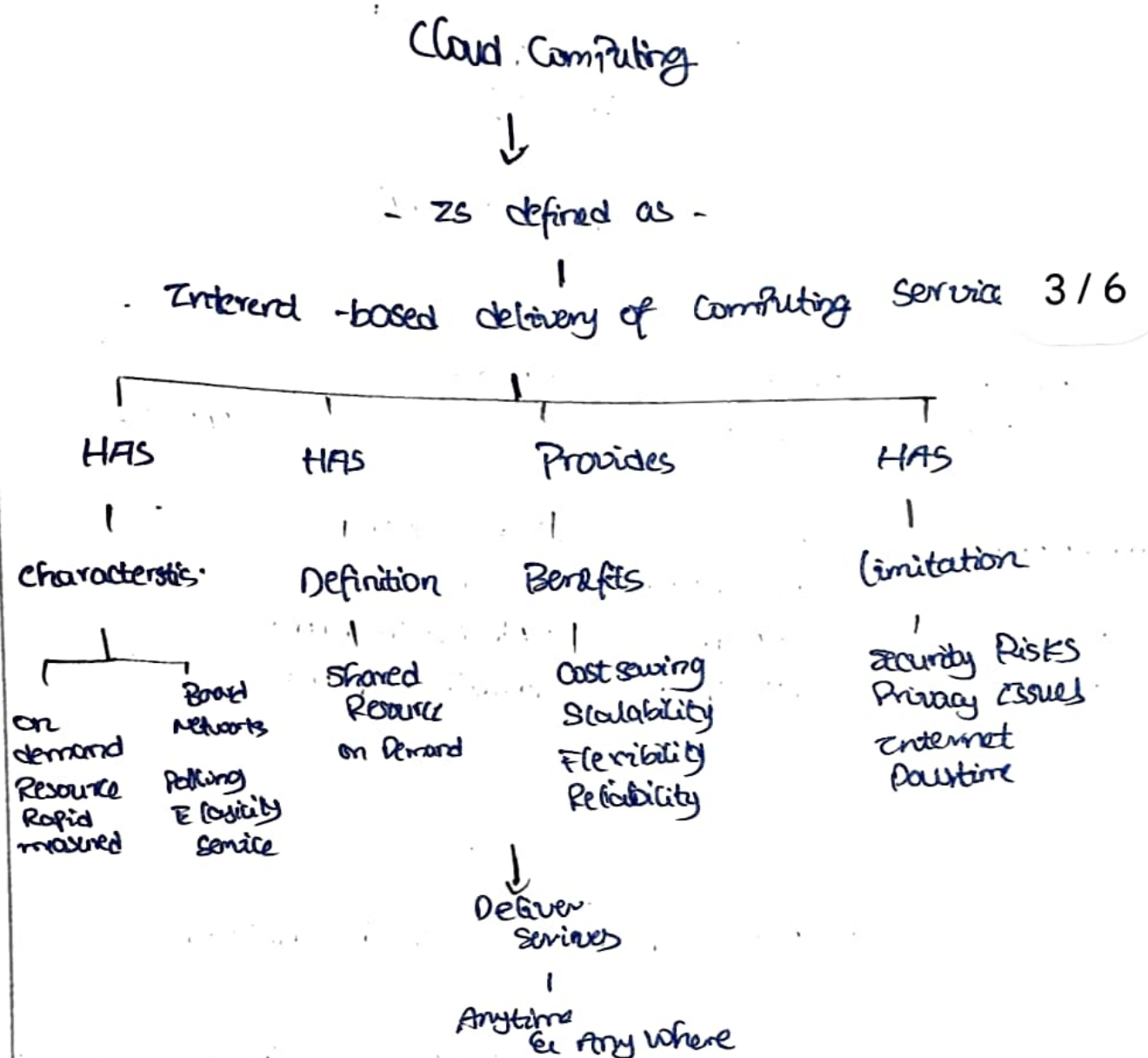
- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or irrelevant	Relationships are mostly incorrect
Hierarchy and Organization	28%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Some structure with weak hierarchy	Not logical organization
Functional Relationships	28%	Most strongly reflect a clear concept	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	14%	Clear, readable, and easy to interpret	Generally readable	Somewhat cluttered	Partly prevented

Concept map :- Cloud Computing Fundamentals, include definition, characteristics, benefits and limitation.



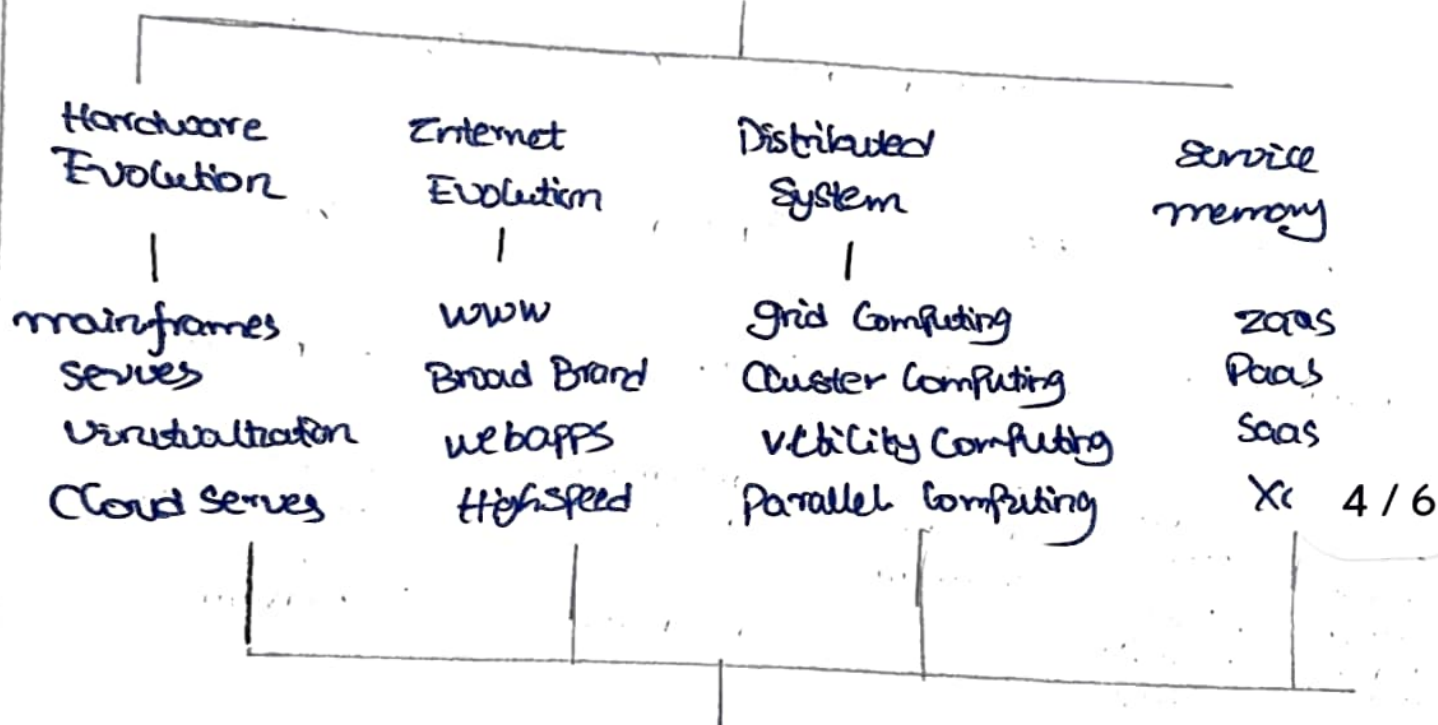
* Cloud Computing is defined as the Internet Based Delivery of computing services.

2)

Concept maps: Evolutions of cloud computing, Connect Home Evolution, internet Software Evolution, distributed Systems and service delivery

Evolutions of Cloud Computing

Developed through



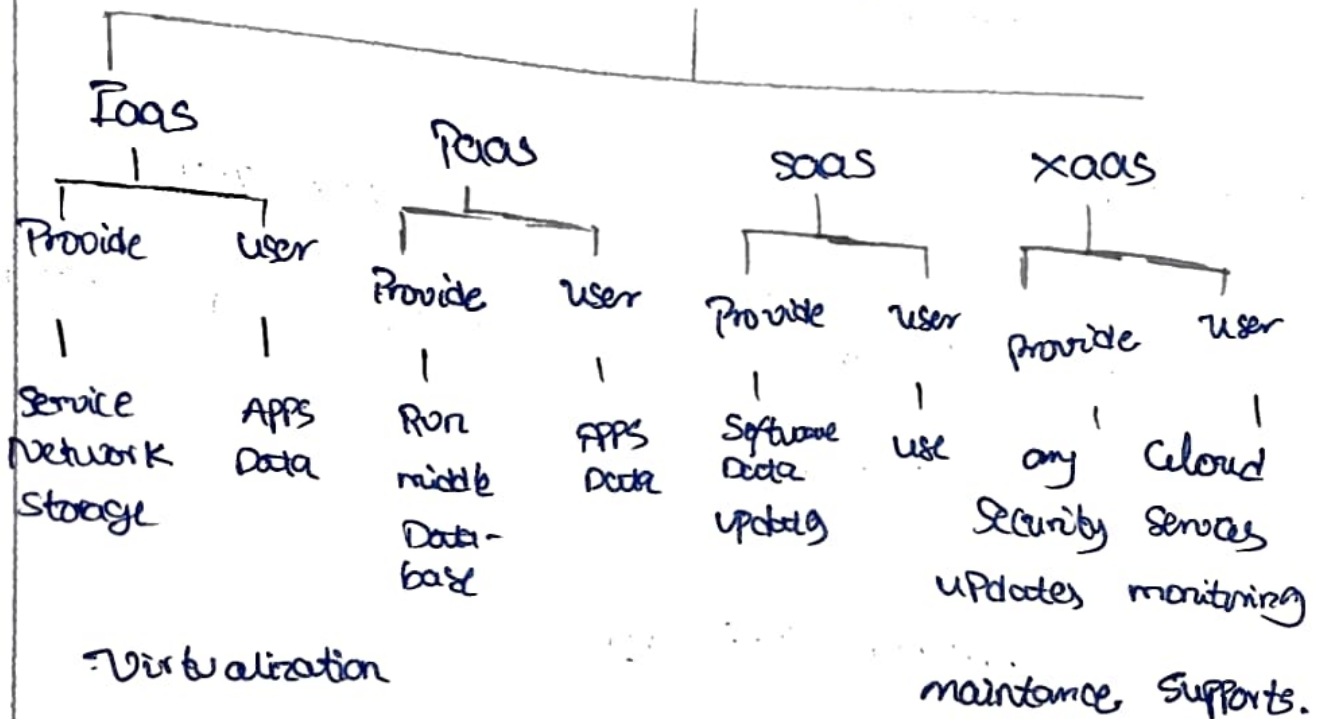
Together enabled Cloud Computing

* Cloud Computing of the Distributed From them so, they can be used From service Delivery, Hardware Evolution, Internet Evolution, Distributed system.

Concept map 3: Service models map .. ZaaS, PaaS, SaaS, and XaaS with Provider responsibilities and User responsibilities

Cloud services model

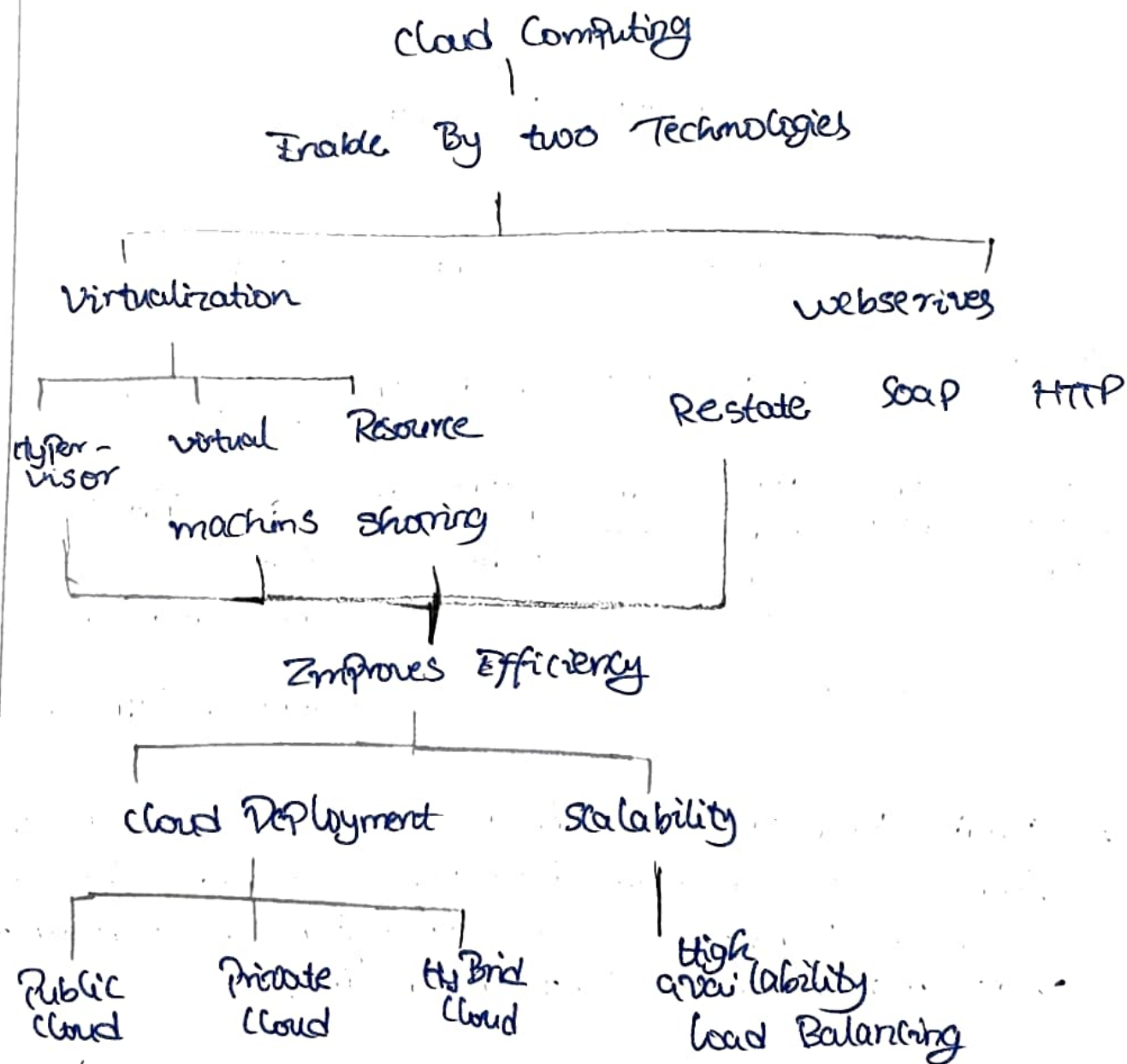
Includes Four types



* Cloud Computing Evolved through advancements in hardware, the internet, distributed system and services and oriented technologies and maintenance, and mainframes, virtualization and Broadband Line.

* PaaS (Platform as a service offers a development, the Environment, software as a service (SaaS) delivery ready to use applications and XaaS includes Cloud based services.

- A) Concept map 4 : virtualization and web services as Enablers of Cloud, show how these two concepts support cloud deployment and scalability



* virtualization and web services are the key technologies that enables Cloud Computing. virtualization allows multiple virtual machines to run single physical server, APIs, REST and SOAP deployment integration.